# Is there a cure for dementia?

There is currently no cure for dementia. In fact, because dementia is caused by different diseases it is unlikely that there will be a single cure for dementia.

Research is aimed at finding cures for dementia-causing diseases, such as Alzheimer's disease (Link: www.nhs.uk/conditions/alzheimers-disease/), frontotemporal dementia (Link: www.nhs.uk/conditions/frontotemporal-dementia/) and dementia with Lewy bodies (Link: www.nhs.uk/conditions/dementia-with-lewy-bodies/).

Read more about the causes of dementia (Link: www.nhs.uk/conditions/dementia/about-dementia/causes/)

Huge strides have been made in understanding how different diseases cause damage in the brain and so produce dementia. And with increased funding over the past few years, there are now many more research studies and clinical trials taking place.

Although a cure may be some years away, there are some very promising advances.

Here are some of the areas researchers are working on and their findings so far.

## Stem cells and dementia

Stem cells are building block cells. They can develop into many different cell types, including brain or nerve cells.

Scientists have taken skin cells from people with certain types of dementia, such as Alzheimer's disease, and reprogrammed them into stem cells in the lab. They've then triggered these stem cells to become brain cells.

By studying these cells, scientists have gained important insights into how the damage to the brain begins and how it might be halted.

These brain cells can also be used to test potential treatments at a very early stage.

## Immunotherapy

Immunotherapy involves boosting the body's immune system to fight disease. It's been effective at treating some other long-term diseases such as cancer.

Different types of immunotherapy treatment for dementia have been trialled in recent years, or are currently being studied. For example, some studies have tried using a vaccination against abnormal proteins that build up in the brain in Alzheimer's disease. Other studies have used monoclonal antibodies (artificial versions of immune system antibodies) to target the abnormal proteins to try to slow the disease progressing.

These studies have had mixed results so far, with some being unsuccessful. But some monoclonal antibody medicines have shown promising results, and are now being considered as treatments for Alzheimer's.

Read more about the specific trials at Join Dementia Research (Link: https://www.joindementiaresearch.nihr.ac.uk/)

Another area being explored by researchers involves specialised immune cells in the brain called microglia. These cells are involved in clearing out debris from the brain.

In Alzheimer's disease, these immune cells appear to become overactive, which may be causing further damage to the brain. Current studies are trying to identify how to prevent this.

### Gene-based therapies

There is great interest in using gene-based therapies to target genes that can cause dementia, such as Alzheimer's disease or frontotemporal dementia.

These gene-based therapies are also being used to reduce the production of proteins involved in a dementia-causing disease, such as tau in Alzheimer's disease.

## Repurposing medicines

Developing new medicines to treat dementia takes many years and millions of pounds.

Repurposing existing drugs used for other conditions is another, often quicker, way of finding medicines to treat dementia.

Current medicines being explored as possible treatments for Alzheimer's and vascular dementia include those used for:

- type 2 diabetes
- high blood pressure
- erectile dysfunction

Find out more about Alzheimer's Society Drug Discovery programme (Link: https://www.alzheimers.org.uk/research/our-research/drug-discovery)

## Identifying who's at risk of dementia

Experts know that damage to the brain caused by Alzheimer's disease can start many years before symptoms appear. If people at risk of Alzheimer's could be identified at an early stage, it is hoped that treatments could be offered that would slow down or even stop the disease.

A major study, called PREVENT (Link: http://news.joindementiaresearch.nihr.ac.uk/prevent-study/), concentrates on people in their 40s and 50s to identify those who are at greater risk of developing Alzheimer's (based on family history and genetics). It aims to understand what is happening in their brains before symptoms appear.

Specialised brain scans, known as PET scans (Link: www.nhs.uk/conditions/pet-scan/), have been developed to study two proteins (amyloid and tau) in the brains of those with Alzheimer's disease. The aim is to increase the understanding of the disease process, and also to identify those people who will benefit most from new drug treatments.

Although PET scans are sometimes used to help with a dementia diagnosis, these highly specialised scans are usually only available as part of clinical trials.

A number of different trials are now under way in people who are currently well but are at increased risk of Alzheimer's disease.

## Prevention is important

Even if we find an effective cure for dementia, it would be better to try to prevent it happening in the first place.

Research has shown that the risk factors for heart disease and stroke – such as raised blood pressure, diabetes, obesity and smoking – are also risk factors for dementia. By modifying or changing these risk factors in mid-life, the risk of dementia could be reduced by up to 30%.

Find out more about risk factors for dementia (Link: www.nhs.uk/conditions/dementia/about-dementia/prevention/)

## Join dementia research

There are dozens of dementia research projects going on around the world, many of which are based in the UK.

If you have a dementia diagnosis or are worried about memory problems, you can help scientists understand more about the disease, and develop future treatments, by taking part in research.

If you are a carer, you can also take part as there are studies into the best ways to care for someone with a dementia diagnosis.

You can sign up to take part in trials on the NHS Join Dementia Research website (Link: https://www.joindementiaresearch.nihr.ac.uk/).

Sign up for NHS Dementia Information Service emails (Link: www.nhs.uk/conditions/dementia/care-and-support/dementia-information-service/)

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Help change lives with research

**130 dementia studies in the UK** are currently looking for people like you to take part.

Find studies now

### More in About dementia (Link: www.nhs.uk/conditions/dementia/about-dementia/)

What is dementia (Link: https://www.nhs.uk/conditions/dementia/about-dementia/what-is-dementia/)

Worried someone has dementia? (Link: https://www.nhs.uk/conditions/dementia/about-dementia/worried-someone-has-dementia/)

Causes of dementia (Link: https://www.nhs.uk/conditions/dementia/about-dementia/causes/)

Can dementia be prevented? (Link: https://www.nhs.uk/conditions/dementia/about-dementia/prevention/)

What are the treatments for dementia? (Link: https://www.nhs.uk/conditions/dementia/about-dementia/treatment/)

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